

**Developing and Validating Annotation Guidelines
for Group-Related Crime Frames in German
Newspapers:
A Comparison of Human Annotation and Large Language
Models**

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Racist discourse in news coverage often emerges cumulatively through recurring framing choices, including whether ethnic, national, or religious groups are associated with crime, danger, or security threats. This project examines whether large language models (LLMs), guided by iteratively refined natural-language annotation instructions, can reliably and validly identify such context-dependent framing in German news. The annotation procedure was developed for a dataset of 659,895 paragraphs from 60 outlets published in 2022 and covering 78 groups. Grounded in Freudenthaler, Ludwig, and Müller (2025)’s framework for measuring fragmented racism, the procedure is intended to support large-scale comparisons of group portrayals across different types of news outlets.

The original label scheme distinguished no crime frame, crime frame positive and crime frame negative cases. The annotation instruction addressed recurring questions in crime reporting, including suspicion before conviction, actor attribution, war reporting, contested accusations, and the relevance of foreign incidents to Germany. Human intercoder reliability increased from Krippendorff’s $\alpha = .560$ in the first oversampled development batch to $\alpha = .786$ in the second batch. In the final random validation sample of 600 paragraphs, it achieved an accuracy of .987, a macro- F_1 of .920, and Cohen’s $\kappa = .839$ against the human consensus labels.

The results indicate that systematic instruction refinement and hardcase review can support reliable LLM-assisted annotation of subtle group-related crime frames. However, the small sample as well as the unbalanced distribution with a lack of positive cases requires further validation on larger and more diverse samples.

1. Introduction

News media do not simply report on ethnic, national, and religious groups; they select which aspects of a group’s presence to foreground, and threat is one of the most consequential of these selections. When a newspaper links a group to crime, danger, or violence, it activates what Entman (1993) calls: *To frame is to select some aspects of a perceived reality and make them more salient in a communicating text, in such a way as to promote a particular problem definition, causal interpretation, moral evaluation, and/or treatment recommendation for the item described.*

This project operationalizes and measures one such frame, *security-crime frame*, understood as the degree to which a news paragraph portrays a group as a source of criminal or security threat (negative valence) versus a contributor to safety, protection, or crime prevention (positive valence), versus containing no such frame at all. We build this measurement on top of Freudenthaler, Ludwig, and Müller (2025)’s recently proposed framework for measuring fragmented traces of racism in large-scale news corpora, which places threat and benefit framing alongside five other sub-dimensions of racist and anti-racist discourse (social position, entitlement, essentialism, homogenization, and othering)

as one of the elements that link ideology to structural inequality and audience attitudes (see Section 2).

Two practical problems make this measurement difficult at scale. First, the underlying corpus is large: our dataset comprises 659,895 German-language news paragraphs from 60 outlets published in 2022, mentioning 78 distinct ethnic, national, or religious groups, far too many for exhaustive human coding. Second, framing is a subtle, context-dependent judgment that simple keyword counting cannot capture reliably: the same lexical item (e.g. “crime,” “police,” “war”) can appear in paragraphs that clearly may or may not frame a group as a security threat, depending on whether the group is cast as perpetrator, victim, bystander, or unrelated third party. We therefore rely on large language models (LLMs) prompted with detailed natural-language coding instructions, following the human-LLM hybrid annotation. Therefore, we (1) draft instructions, (2) test and improve their coverage, (3) validate against a curated sample of hard and soft cases, and (4) validate against a representative, independently human-coded sample.

The broader goal of the project is to operationalize ethnic and racial bias in news communication and to enable systematic comparisons of how different groups are portrayed across German news outlets. Within this broader framework, our research question is: *Can large language models, guided by iteratively refined natural-language annotation instructions, reliably and validly identify group-related crime framing in German news coverage, and how does their performance compare with that of human annotators?*

Because this context-dependent judgment cannot be reduced to simple keyword matching, answering the question requires more than a single accuracy measure. We therefore document, at each stage of instrument development, how reliably human annotation agrees with each other and how validly LLM annotations track that human judgment, so that the resulting instrument’s performance can be trusted rather than merely asserted.

The remainder of this report is organized as follows. Section 2 situates the security-crime frame within Freudenthaler, Ludwig, and Müller (2025)’s broader framework and the wider framing-effects literature. Section 3 describes the corpus, its outlets, and the preprocessing steps. Section 4 documents the sampling strategy, the four-stage annotation pipeline, the models used, and the two side-experiments we ran (LLM ensembling and confidence-based routing). Section 5 reports intercoder reliability, LLM validity, the ensembling and confidence-routing comparison, and the substantive framing patterns we find. Section 6 discusses what these results mean, Section 7 discusses limitations and future work, and Section 8 concludes.

2. Theoretical Background

2.1. Racism as a Fragmented Discourse Pattern

Racism research distinguishes structural inequality from the ideological, attitudinal, and behavioral processes that maintain it (Bonilla-Silva & Baiocchi, 2001; Fredrickson, 2002; Omi & Winant, 2014); news production is one of the arenas where this ideology is reproduced or contested (Feagin, 2020; van Dijk, 1991). Yet computational approaches to measuring racism in text typically operationalize a single, narrow indicator rather

than a multidimensional construct (Kathirgamalingam et al., 2025). (Freudenthaler, Ludwig, & Müller, 2025) address this gap with a six-dimension framework (social position, entitlement, essentialism, homogenization, othering, and threat/benefit framing) that places a group’s portrayal on a spectrum from overtly racist to egalitarian discourse, rather than judging individual texts as racist or not.

This project isolates and operationalizes the sixth dimension, threat and benefit framing, narrowed further to a single frame family: the *security-crime frame*, in line with Freudenthaler, Ludwig, and Müller (2025)’s own observation that full multidimensional measurement is rarely feasible within a single study, and because this frame has an established place in the framing-effects literature as one of the more behaviorally consequential frames applied to migrant and minority groups (Section 2.2).

2.2. Framing Theory and the Security-Crime Frame

(Entman, 1993) defines framing as the selective emphasis of some aspects of an issue over others to promote a particular interpretation. The security-crime frame casts a group’s presence primarily in terms of danger, criminality, or eroded public safety, as opposed to frames emphasizing economic contribution, cultural difference, or humanitarian circumstances (Eberl et al., 2018). Threat-framed coverage measurably shifts audience attitudes and emotions toward immigrants and minority groups, and these effects are frame-specific rather than uniform across news content (De Vreese et al., n.d.; Guo et al., 2023; Lecheler et al., 2015), which is why frame valence, threat versus protection, and not merely its presence, is the quantity worth measuring.

Crime framing, as we use the term, refers to how media texts associate a group with crime, danger, security, or prevention, thereby shaping readers’ perceptions and contributing to bias. In the literature this valence is typically labeled *threat* and *benefit* framing (Eberl et al., 2018; Freudenthaler, Ludwig, & Müller, 2025); in the remainder of this report we instead refer to *negative* and *positive* crime framing, to keep the terminology anchored to the specific frame family we operationalize rather than to the broader threat/benefit dimension of Freudenthaler, Ludwig, and Müller (2025)’s six-dimension framework. We focus this measurement on Germany specifically because it is one of the European media systems most extensively documented in the immigration-discourse literature (Eberl et al., 2018), letting us situate our findings against an established base of comparative evidence rather than a national case with no prior benchmark.

Translating this conceptual definition into a reproducible coding decision for hundreds of thousands of paragraphs requires explicit annotation instructions that resolve boundary cases a definition alone leaves open, such as suspicion versus conviction, or war reporting versus crime reporting; these instructions are what both our human coders and the LLM are given to produce a label. The full operational definition is given in Section 4 and reproduced verbatim in Appendix A.1.

2.3. LLMs as Content-Analysis Coders

Two measurement properties are central to evaluating any coding scheme, whether applied by a human or a model. *Reliability* asks whether independent coders applying the same instructions to the same material arrive at the same label; we quantify this using Krippendorff’s α , a chance-corrected agreement statistic for nominal-category content analysis (Neuendorf, 2017) that ranges from -1 (systematic disagreement) to 1 (perfect agreement), with 0 indicating agreement no better than chance. *Validity* asks whether a coder’s labels track an external, trusted ground truth rather than merely agreeing with each other; we quantify this using Cohen’s κ , a similarly chance-corrected measure of agreement between LLM annotations and human consensus labels. Both statistics correct for the possibility that two coders, or a coder and a model, might agree simply by chance, which raw percentage agreement does not.

A growing literature asks whether LLMs can substitute for, or assist, human coders in content analysis. Van Atteveldt et al. (2021) find only human coders reliably exceed Krippendorff’s $\alpha = 0.8$ on sentiment annotation, with dictionary ($\alpha \approx 0.34$) and deep-learning ($\alpha \approx 0.5$) methods falling short. MacPhail et al. (2016) and O’Connor and Joffe (2020) treat persistent intercoder disagreement as a diagnostic signal that a concept is not yet clearly defined, rather than a technical checkbox; iterating instructions in response is itself part of a defensible measurement procedure. Jumle et al. (2025) show framing categories remain a harder automated-coding target than simpler sentiment or topic labels, and Weber and Reichardt (2024) find generative LLMs cannot be used for precise annotation “out of the box” regardless of whether one fine-tunes a classifier or prompts directly: careful validation and tailored prompting are required either way. This is the backdrop for our four-stage pipeline (Section 4), in which we treat both human and LLM annotation as processes whose reliability must be measured, diagnosed, and iteratively improved, and we report each stage of that process rather than only the final numbers.

3. Data

Our dataset consists of 659,895 German-language news paragraphs published by 60 online news outlets in 2022. Articles were collected, via a web crawler targeting reportings about nationality, migrant, ethnic, and religious groups, then segmented into paragraphs, which serves as our unit of analysis, deduplicated, and dropping paragraphs under 15 words because they are claimed as too short to support a framing judgment.

Each paragraph contains a masked group of interest, represented as [Gruppe 1], while references to other groups are replaced with [Andere Gruppe]. This masking procedure was used so that both human annotators and the LLM would base their judgments on the explicit content of the paragraph rather than on prior assumptions or stereotypes about a named nationality, migrant, ethnic, or religious group. This is particularly important for measuring potentially discriminatory crime framing, since the aim is to identify cumulative discourse patterns without reproducing or endorsing the frames under study. The original group identity is retained separately in the `group` column and reattached after annotation for the downstream analysis.

The data contains 78 distinct nationality, migrant, ethnic, or religious groups, which can be divided into two broad categories. The first and larger category consists of country-of-origin nationalities, such as Ukrainians (UKR), Russians (RUS), French people (FRA), Turkish people (TUR), and Afghans (AFG). The second, more heterogeneous category contains religious, ethnic, migration-related, and other cross-national groups, including refugees (REFUG), migrants (MIGR), Muslims (MUSL), Jewish people (JUDE), Kurds (KURD), and asylum seekers (ASYL).

This distinction is important for interpreting the group-level comparisons in Section 5.4. Broad categories such as REFUG and MIGR aggregate coverage across multiple nationalities and are therefore more likely than specific nationality groups to appear in general discussions of migration policy and security. Table 1 lists the ten most frequently represented groups.

Group	Code	Paragraphs
Refugees	REFUG	117,132
Ukrainians	UKR	72,028
Russians	RUS	70,398
British people	GBR	37,580
Americans	USA	35,357
Hungarians	HUN	34,563
French people	FRA	27,939
Migrants	MIGR	25,207
Jewish people	JUDE	24,779
Spanish people	ESP	15,249

Table 1: The ten most frequently represented groups in the full dataset.

The news outlets are classified into four categories (Table 2): mainstream national outlets (e.g. *Süddeutsche Zeitung*, *FAZ*, *Spiegel*, *Bild*), regional outlets (e.g. *Tagesspiegel*, *Merkur*, *Weser Kurier*), and two categories of alternative media distinguished by political orientation: alternative right (e.g. *Epoch Times*, *Compact*, *PI News*) and alternative left (e.g. *Junge Welt*, *Telepolis*, *Nachdenkseiten*). In summary, Mainstream and regional outlets together account for approximately 90 percent of all paragraphs in the dataset.

Outlet category	Outlets	Paragraphs	Share
Mainstream national	14	333,782	50.6%
Regional	17	259,678	39.4%
Alternative right	16	41,186	6.2%
Alternative left	13	25,249	3.8%
Total	60	659,895	100.0%

Table 2: Dataset composition by outlet category.

4. Method

4.1. Sampling Strategy

Manually annotating the full dataset was infeasible. The annotation workflow therefore relied on a series of smaller samples drawn from the full dataset. These samples served different purposes: initial exploration, instruction development, hard-case testing, and final validation.

As a first exploratory step, 2,000 paragraphs were annotated from the LLM using an initial, simple version of an annotation instruction by Freudenthaler, Müller, and Ludwig (2025), for group-related security, threat, and benefit frames. This preliminary LLM annotation showed that the three labels were unevenly distributed. While 57.26 percent of the paragraphs were labeled `NO_CRIME_FRAME` and 41.89 percent were labeled `CRIME_FRAME_NEG`, only 0.85 percent received the label `CRIME_FRAME_POS`. Thus, negative crime framing was sufficiently present for instruction development, whereas positive crime framing was extremely rare. A fully balanced sample across all three classes was therefore not feasible.

To ensure that the annotation instruction was nevertheless tested on a sufficient number of relevant and difficult cases, we constructed an oversampled sample of 500 paragraphs. This sample combined cases from the preliminary 2,000-case annotation with additional paragraphs drawn from the full dataset. The sample was designed to include approximately 40 percent likely crime-frame paragraphs, drawing on all available positive cases and filling the remaining share with negative crime-frame cases; 30 percent hard cases; 25 percent likely non-crime paragraphs; and 5 percent keyword-based cases. The keyword search used terms associated with crime and security discourse, including *Diebstahl*, *Prävention*, *Terror*, and *Abschiebung*. The hard cases were identified by annotating each paragraph ten times and calculating agreement across the repeated runs. Paragraphs with less than 80 percent agreement were stored in a separate hardcase file, from which the hard-case share of the sample was drawn.

The 500-case development sample was then divided into two batches of 250 paragraphs. The first batch was used for a first round of instruction refinement, especially regarding the distinction between crime framing and factual reporting on war, armed conflict, or military violence. The second batch was used for a further refinement of the coding instructions, including the decision to restrict crime and security frames to cases with a clear connection to Germany. These revisions are described in more detail in the following section 4.2.

The oversampling strategy was used only for instruction development and hard-case testing. As the annotation instructions were revised, the final label distribution no longer corresponded to the originally intended sampling proportions. The oversampled development sample should therefore not be interpreted as a representative estimate of the prevalence of crime framing in the full dataset.

After the annotation instructions had been finalized, model performance was evaluated on an independent random sample of 600 paragraphs. It provides a more realistic test of whether the model could identify crime frames under the naturally imbalanced conditions

of the dataset. In this final random sample, only approximately 4 percent of paragraphs were related to crime, making it a "stress test" for both human-model agreement and the model's ability to detect relatively rare crime-frame cases.

4.2. Annotation Instruction Steps

We followed Freudenthaler (2024)'s four-stage routine for developing and validating an LLM-based content-analysis instruction. Because the accompanying `bacchuss` package is available only in R, whereas our workflow was implemented in Python, we reproduced the logic and analysis of the routine as closely as possible within our Python workflow. The final validation, based on an independent random sample, was preceded by an initial, non-representative phase involving the instruction drafting, testing, and revision of the annotation instruction. During this development phase, we drafted the annotation instruction, tested the coverage on smaller samples, and identified hardcases through human annotation disagreements and repeated LLM runs. Coverage problems and recurring error patterns identified at each stage lead to multiple revisions and the selection of few-shot examples used during the development rounds. When we, as human annotators, disagreed on a case, we reviewed it together and agreed on a final label. These consensus labels then served as the human reference labels for the LLM. In Steps 1–3 we developed and refined the annotation instruction, while in Step 4 we applied the finalized instruction using the selected Minstral model on an independent random sample which was annotated by both humans and the LLM.

Step 1: Drafting an Annotation Instruction

The first step consisted of drafting an annotation instruction for our LLM-based annotation. Following Freudenthaler (2024), the instruction contained five main elements: a role description, context information about the data, an explanation for each label, decision rules, and a fixed output format.

The role description instructed the model to act as an expert in content analysis for a study on the media representation of ethnic and social groups in German news. The context section explained that the model would receive German newspaper paragraphs in which group identities were masked. The focal group was represented as `[Gruppe 1]`, while other groups were represented as `[Andere Gruppe]`. The model was instructed to focus only on the representation of `[Gruppe 1]` to decide on a label and to treat other groups as context only.

We drafted a three-class coding scheme: `NO_CRIME_FRAME`, `CRIME_FRAME_NEG`, and `CRIME_FRAME_POS`. Although negative and positive crime framing are conceptually distinct, the LLM was instructed to assign exactly one of these three labels to each paragraph.

The label `NO_CRIME_FRAME` was assigned when the paragraph did not explicitly link `[Gruppe 1]` to crime, security, danger, suspicion, violence, terrorism, prevention, or protection. The label `CRIME_FRAME_NEG` was assigned when `[Gruppe 1]` was portrayed as a source of crime, danger, violence, suspicion, terrorism, or security threat. This included cases in which members of `[Gruppe 1]` were described as perpetrators, suspects,

accused persons, or sources of public or national insecurity. The label `CRIME_FRAME_POS` was assigned when a crime- or security-related frame was present and `[Gruppe 1]` was portrayed as actively contributing to safety, protection, crime prevention, or de-escalation. Examples include prevention measures, social work, deradicalization, policing, or other activities through which members of `[Gruppe 1]` contribute to public safety. An example of a paragraph labeled as positive crime frame can be found in the Appendix A.3.

The decision rules were designed to keep model hallucination small by requiring the model to decide only on what is written in the paragraph itself. First, a crime or security frame had to be explicitly connected to `[Gruppe 1]`; if crime, violence, danger, or prevention was mentioned only in relation to another group, the paragraph was coded as `NO_CRIME_FRAME`. Second, the model was instructed not to infer criminality, danger, or prevention from background knowledge about a nationality, religion, migrant group, or political context, crime-related vocabulary alone was not sufficient; what mattered was whether the paragraph itself framed `[Gruppe 1]` as criminal, dangerous, or, conversely, as contributing to safety.

Two further rules addressed specific edge cases that we implemented from the annotation instruction by Freudenthaler, Müller, and Ludwig (2025). One is that undocumented migration status alone, e.g., “illegal immigrants”, did not lead to `CRIME_FRAME_NEG` unless a concrete offense was named, whereas describing a group member as a smuggler (“Schleuser”) did count, since smuggling is itself a criminal offense. And the label did not require a judicial conviction: suspicion, ongoing investigation, and unconfirmed allegations counted once explicitly linked to `[Gruppe 1]` (e.g., “mutmaßlich”/allegedly, “verdächtig”/suspected, “Ermittlungen gegen”/investigation against), reflecting how crime reporting typically concerns suspects rather than convicted offenders.

For ambiguous cases that couldn’t be resolved by these rules, the instruction used a reader-oriented fallback heuristic. The model was asked whether a typical reader would perceive `[Gruppe 1]` as criminal, suspicious, security-threatening, or as contributing to safety or prevention. If the paragraph did not make such a reading salient, the model was instructed to default to `NO_CRIME_FRAME`. This fallback links the annotation task to the broader framing concept of the project: the relevant question is not whether a group is objectively dangerous, but whether the paragraph makes crime or security salient in relation to that group.

Finally, we specified a fixed output format. The model was required to return exactly one of the three valid labels. In addition to the main instruction, we used a short reminder after each paragraph that repeated the central rule: assign a crime-frame label only when `[Gruppe 1]` is explicitly connected to crime, security, danger, violence, suspicion, prevention, or protection. Because smaller LLMs can produce incorrect labels or additional explanatory text instead of the exact label string. To avoid incorrect matches caused by overlapping label names, the parser checked for `NO_CRIME_FRAME` before checking for the two crime-frame labels.

At this stage, the instruction provided a first working structure and operational definition of group-related crime framing based on the existing distinction between security threat and benefit. More detailed rules for recurring boundary cases were introduced during the coverage and hardcase testing in Step 2 and the subsequent instruction revisions in

Step 3. These included rules for war reporting, fictional and satirical content, propaganda and contested claims, and the requirement of an explicit connection to Germany.

Step 2: Coverage and Hardcase Testing

In the second step we tested whether the draft instruction covered the concept broadly enough and whether the model applied the labels consistently across repeated runs. Following the logic of Freudenthaler (2024), this step was not intended to estimate final model performance. Instead, it was used to identify coverage problems, ambiguous cases, and recurring error patterns that could be addressed through more explicit decision rules.

Given the fact that we used Python as our coding platform, we implemented the same logic of the `briseus` function from the `bacchuss` package. The annotation instruction and model settings were stored in a separate configuration file, so that the same instruction could be accessed consistently across the different annotation scripts and notebooks. This also made it easier to revise the instruction iteratively while keeping the annotation pipeline reproducible.

As already described in Section 4.1, the hardcases used in the development sample were identified through repeated LLM annotation. We tested the draft instruction on samples of 20, 200, and later 2,000 paragraphs, annotating each paragraph ten times at a temperature of $T = 0.7$. Paragraphs with less than 80% agreement across runs were classified as LLM hardcases, stored separately, and manually reviewed to identify ambiguities in the annotation instructions.

The hardcase review showed that several types of cases required clearer decision rules. One recurring issue concerned the relation between [Gruppe 1] and the reported crime. Some hardcases mentioned crime, violence, terrorism, or security threats, but linked these elements to another perpetrator, another group, victim or a broader political situation rather than explicitly to [Gruppe 1]. We therefore strengthened the rule that a crime frame can only be assigned when [Gruppe 1] is explicitly connected to criminality, suspicion, violence, terrorism, danger, prevention, or protection. If another group, actor or victim is related to a crime the paragraph receives `NO_CRIME_FRAME`.

One typical hardcase involved reporting on protests in Iran. The paragraph first stated that the pro-government news agency Tasnim presented the situation differently and then described protesters as “bewaffnete Terroristen” who had allegedly set buildings on fire. The word “differently” immediately distanced the paragraph from the accusations and presented them as propaganda rather than as verified facts. With that in mind, [Gruppe 1] is not seen as a source of crime or threat and was labeled as `NO_CRIME_FRAME`. The LLM struggled to recognize these subtle nuances that influenced the labeling decision. The full paragraph can be found in Appendix A.4.

A second recurring issue was when to assign a negative crime frame. The hardcases showed that the instruction needs to make it more clear that Group 1 does not have to be convicted to be associated with a crime. Suspicion, investigations, allegations, police searches, arrests or wording such as “mutmaßlich”, “verdächtig”, “beschuldigt”, or “soll getan haben” should lead to a negative crime framing. This is an important distinction because it is common in crime reporting, where people are often described as suspects

before a court has established guilt.

A third problem we identified in the hardcases concerned demonstrations and protests. During the first rounds of testing, the LLM sometimes labeled demonstrations or political assemblies as `CRIME_FRAME_POS` even when they were not connected to policing, violence, prevention, or public safety. We therefore added an explicit rule that demonstrations, protests, and political assemblies should normally be coded as `NO_CRIME_FRAME`. They were assigned `CRIME_FRAME_NEG` only when [Gruppe 1] was explicitly linked to violence, criminal acts, riots, threats, or security risks. They were assigned `CRIME_FRAME_POS` only when they led to concrete legal, political, policing, or preventive changes, or were presented as contributing to greater safety, protection, prevention, or the rule of law.

A fourth issue concerned cases that appeared positive because they involved security measures, but did not actually portray [Gruppe 1] as contributing to safety. For example, deportation or restrictive migration measures presented as a way to reduce crime should not be coded positive. If in such cases, the group or migration is associated with as a source of crime, the appropriate label is `CRIME_FRAME_NEG` if not `NO_CRIME_FRAME`. Based on these observations, we systematically updated the annotation instruction.

Step 3: Validation Against a Curated Non-Representative Sample

Once coverage looked broad, we assembled a non-representative test set of 500 paragraphs mixing hard cases identified in Step 2, clear examples of each label, and additional keyword-searched cases, and we both independently hand-label a first batch of 250 of these paragraphs to serve as a human "ground truth". Model annotation (`minstral-3-14b`, university API, temperature 0) against this human standard achieved limited agreement, evaluated using macro- F_1 (the unweighted average of per-class F_1 , chosen over accuracy or micro-averaged measures specifically because it does not let a dominant class mask poor performance on rarer ones; (Sokolova & Lapalme, 2009)) and Cohen’s κ (Table 4, row “Step 3”); the 86 disagreement cases were extracted as model hard cases for further instruction refinement, using consensus-coded curated samples for this stage rather than a purely random sample.

Step 3.1 Prompting Strategy Comparison and Instruction Revision

This stage is where prompting strategy was systematically varied, still on the same non-representative 250-paragraph set used in Step 3. We compared three prompting conditions: zero-shot, few-shot with 13 hand-written examples targeting error patterns observed in Step 3, and few-shot with chain-of-thought explanations.

In writing the 13 few-shot examples, we followed established curation principles: an approximately balanced number of examples per label rather than mirroring the dataset’s natural, heavily skewed label distribution; at least one unambiguous example per label to anchor the model to the core concept; and additional examples chosen to illustrate decision rules the model had previously applied incorrectly, such as the victim-versus-perpetrator distinction and the war-reporting exclusion. We wrote these examples ourselves rather than drawing them from the dataset to avoid using evaluation paragraphs in the prompt.

Few-shot prompting outperformed both zero-shot and few-shot with chain-of-thought on this set (Table 4).

We then revised the instructions further. Among other refinements, two separate rules were added in this round. Foreign incidents were coded as `CRIME_FRAME_NEG` only when an explicit connection to Germany existed, narrowing the frame to incidents relevant to German society rather than any crime report mentioning a nationality. We also clarified that deportations are not automatically coded as `CRIME_FRAME_POS`, because they present the deportee as the target of an adverse state action rather than as evidence of a security benefit.

To assess the revised procedure on previously unused data, we independently annotated the second 250-paragraph batch from the same oversampled development sample. Human intercoder reliability, measured with Krippendorff’s α , was higher than in the first batch ($\alpha = .786$ compared with $\alpha = .560$; Section 5.1). Against the resulting human consensus labels, the revised procedure achieved macro- $F_1 = .849$ and $\kappa = .702$ (Table 4). Although this second batch had not been used in the initial prompting comparison, it came from the same oversampled development sample and had a different label distribution. The final random validation therefore provides the more relevant estimate of performance under the natural distribution of the dataset.

Step 3.2: Comparing Different LLMs

Because a single small model’s validity plateaued despite the revised few-shot instructions above, we turned to comparing and combining different LLMs, the activity Freudenthaler (2024)’s routine designates Step 3.4. We tested two ways of using a second, larger model, both evaluated on the same 250-paragraph test set used in Step 3 and the revision round above: first confidence-based routing (cheaper, tried first), then multi-model ensembling (more expensive, tried once routing failed).

Step 3.3: Confidence-Based Routing

The motivation was cost: rather than sending every paragraph through the larger, slower `Llama-3.3-70B` model, we wanted `ministral-3-14b` to flag which paragraphs it was actually unsure about and route only those onward. `ministral-3-14b` rates its own confidence in each label on a 1–5 scale, and paragraphs at or below a confidence threshold of 3 are re-routed to `Llama-3.3-70B` for a second opinion. We dropped this approach because the confidence scores turned out to be uncalibrated, in the wrong direction: accuracy was higher at a supposedly less certain confidence level than at a supposedly more certain one, backwards from what a usable confidence signal would look like. Consistent with this, routing triggered on 244 of 250 paragraphs (97.6%) because the small model rarely reported high confidence at all, and overall performance after routing was worse than `ministral-3-14b` annotating alone without any routing (Table 6). We report this as a negative result: small LLMs cannot be assumed to reliably self-assess their own uncertainty, and that assumption should be tested, not taken for granted, before building a routing pipeline around it.

Step 3.4: Multi-Model Ensembling

Since self-reported confidence failed as an uncertainty signal, we tried a more principled alternative: actual disagreement between two independently annotating models, which is an objective signal rather than a subjective self-assessment. We ran **ministral-3-14b** (Model A) and **Llama-3.3-70B**¹ (Model B) independently on the same test set; where the two models agreed, we kept their shared label; where they disagreed (74 of 250 paragraphs, 29.6%), a third model, **qwen3.5-27b**, adjudicated the conflict by inspecting both labels and their explanations, siding with Model A on 61 of the 74 conflict paragraphs and Model B on 13. This ensemble outperformed single-model annotation against the human consensus labels (Table 5). We nonetheless did not adopt this configuration, for two reasons. First, the improvement over **ministral-3-14b** alone was marginal relative to the cost of running two models plus a tiebreaker on every paragraph. Second, because this is the same test set used to develop the few-shot instructions in the revision round above, not an independently drawn one, even this marginal gain is subject to the overfitting risk discussed in Section 7 and should not be read as a validated generalization estimate. Taken together, both comparisons indicated that the additional complexity of a multi-model pipeline was not justified by the limited performance gain. The final independent validation therefore used a single **ministral-3-14b** model with the finalized instructions in an instruction-only zero-shot setting.

Step 4: Final Representative Validation

After the annotation instructions had been finalized, we conducted a final validation on an independently drawn random sample of 600 paragraphs from the full dataset. As described in Section 4.1, this sample was not oversampled. Unlike the development samples used in Steps 1–3, it was therefore intended to test whether the final annotation procedure also worked under the naturally imbalanced conditions of the dataset.

We labeled the 600 paragraphs independently using the finalized annotation instructions. Cases in which we disagreed were treated as human hardcases. We resolved them through manual reviewing and agreed on one consensus label. The resulting final labels served as the ground truth against which the LLM was evaluated.

For the evaluation, we only distinguished between **NO_CRIME_FRAME** and **CRIME_FRAME_NEG** cases. The **CRIME_FRAME_POS** category was excluded from the final evaluation because it occurred too rarely during the development phase to support reliable classification. In the two 250-paragraph development batches, only one positive crime-frame case was identified after refining the instruction, which was far too sparse to allow for a meaningful validation.

As in the previous hardcase checks, each paragraph was annotated ten times with temperature $T = 0.7$. The final model label for each paragraph was determined by majority vote, that is, by selecting the label that occurred most often across the ten runs. Paragraphs with less than 80 percent agreement across the ten runs were flagged as model hardcases and inspected separately. These hardcases were not used to further

¹Quantized variant **Meta-Llama-3.3-70B-Instruct-AWQ-INT4**, served via the university API.

tune the instruction, but served as an additional diagnostic check of whether the finalized instruction still produced unstable model decisions in the representative sample.

The final model validity was evaluated by comparing the LLM labels with the human consensus labels. In our analysis we looked at accuracy, precision, recall, macro- F_1 , weighted- F_1 , and Cohen’s κ to make any statements regarding the model performance. As a note, human intercoder reliability was assessed before disagreements were resolved and the consensus labels were created.

5. Results

5.1. Human Intercoder Reliability

Table 3 summarizes the human intercoder reliability across the batches double-coded by us, both human annotators. Reliability rose substantially between the first two 250 paragraph batches. Raw percentage agreement increased from 76.8% to 94.4%, while Krippendorff’s α increased from 0.560 to 0.786. Using the interpretation bands common in this literature ($\alpha \geq 0.8$ very good, 0.6–0.8 good, 0.4–0.6 moderate, 0.2–0.4 fair, below 0.2 poor; (Weber & Reichardt, 2024)), this represents an improvement from moderate to good agreement.

This increase shows that the iterative annotation process improved not only the annotation instruction, but also the intersubjective reproducibility of the human coding procedure. After several rounds of hardcase inspection, disagreement discussion, and codebook revision, we developed a more shared understanding of the distinction between NO_CRIME_FRAME and CRIME_FRAME_NEG. In particular, the revised instructions made several recurring boundary cases more explicit, including the required link to [Gruppe 1], the distinction of propaganda, and the Germany-relatedness criterion. The improvement in human reliability therefore provides evidence that the final coding scheme was more precise and easier to apply consistently than the earlier versions. However, the improvement cannot be attributed solely on that, since through the annotations we also gained experience to know what we need to pay attention to and we became more familiar with the instruction and rules. Furthermore, the two batches differed in their label distributions and types of cases, which also cannot be ignored.

The first batch contained a more balanced distribution of NO_CRIME_FRAME and CRIME_FRAME_NEG, whereas 213 of the 250 paragraphs in the second batch were labeled as no crime cases. Raw agreement can increase when one category dominates the sample. Krippendorff’s α is therefore the more informative reliability measure, because it adjusts for agreement expected by chance. The fact that α also rose substantially suggests that the improvement was not solely due to the more imbalanced label distribution.

Raw agreement was also high in the final random validation sample of 600 paragraphs. We disagreed on only 27 cases, corresponding to 95.5% agreement, while Krippendorff’s α was .616, which was lower than in the second development batch. Based on the final human consensus labels, only 26 paragraphs were classified as crime negative, while the remaining 574 were classified as no crime frame. The final validation sample was therefore substantially more imbalanced than the oversampled development batches.

Because agreement on the dominant category was already highly likely, the relatively small number of disagreements involving the rare crime-frame cases had a stronger effect on the chance-corrected reliability score.

The human hardcases in the final validation sample mainly concerned whether the reported crime case had a clear enough connection to Germany. The remaining disagreements were therefore mostly about how to apply the more specific rules added during instruction development. In this sense, the final reliability results show that repeated hardcase review and instruction revision helped us apply the coding scheme more consistently, while also showing which borderline cases remained difficult for human annotators.

Batch	n	% Agreement	Krippendorff's α
First batch (Step 3 / revision test set)	250	76.8%	0.560
second development batch (new250)	250	94.4%	0.786
Step 4: final validation sample	600	95.5%	0.616

Table 3: Human intercoder reliability across annotation batches.

5.2. LLM Validity Against Human Consensus Labels

Table 4 reports LLM agreement with the human consensus labels across instruction refinement and final validation. Overall, the results show that model performance improved substantially as the annotation instructions became more specific, although the different samples and label schemes mean that the scores should not be interpreted as a strictly controlled comparison.

On the first 250-paragraph development batch, the zero-shot condition reached Cohen's $\kappa = .415$ and macro- $F_1 = .563$. Adding few-shot examples improved performance to $\kappa = .638$ and macro- $F_1 = .712$, suggesting that concrete examples helped the model apply the annotation rules to ambiguous cases. Adding chain-of-thought explanations did not provide a further benefit: the few-shot-plus-chain-of-thought condition reached $\kappa = .472$ and macro- $F_1 = .574$. For this sample, concise examples were therefore more effective than longer explanatory prompts.

On the second 250-paragraph development batch, the refined annotation procedure reached $\kappa = .702$ and macro- $F_1 = .849$. This improvement came along with the addition of more precise decision rules, particularly the Germany-relatedness criterion and the exclusion of war-related reporting. These rules likely made several previously ambiguous cases easier to classify. However, because this batch was still part of the curated and oversampled development sample, the result should not be treated as a representative estimate for the full dataset.

The strongest and most relevant validity estimate comes from the final random sample of 600 paragraphs. On this sample, the model reached a macro- F_1 of .920 and Cohen's $\kappa = .839$, disagreeing with the human consensus labels in only eight cases. These results indicate strong agreement beyond the dominant NO_CRIME_FRAME category. However,

only 26 paragraphs were classified as `CRIME_FRAME_NEG`, so conclusions about the model’s performance on the minority class are based on relatively few cases and should be interpreted cautiously.

The final scores are not directly comparable to the development-stage scores without qualification. The development samples were curated and oversampled and initially used a three-class scheme, whereas the final validation used a random sample and focused only on the distinction between `NO_CRIME_FRAME` and `CRIME_FRAME_NEG`. The higher final scores may therefore reflect both the more precise annotation instructions and the restriction of the validity evaluation to two categories. Because the samples and label distributions also differed, the influence of these factors cannot be determined separately.

Across the ten repeated runs, 14 paragraphs received less than 80% agreement and were therefore classified as LLM hardcases. These cases mainly concerned the distinction between perpetrators and victims, as well as the application of the Germany-relatedness criterion.

The eight disagreements with the human consensus labels showed a similar pattern. The model sometimes failed to determine whether [Gruppe 1] was presented as a perpetrator, victim, suspect, or only indirectly connected to the event. Other disagreements concerned whether foreign or EU-related events had a sufficient connection to Germany.

One illustrative error involved a report about a fatal stabbing in which [Gruppe 1] was most likely presented as the victim rather than the perpetrator, but the model assigned `CRIME_FRAME_NEG`. This indicates that the LLM still had difficulties with more complex paragraphs in which the roles of the victim and perpetrator were not immediately clear. The complete example is provided in Appendix A.5.

Overall, few-shot prompting performed best in the initial prompting comparison, while the finalized zero-shot procedure achieved strong agreement with the human consensus labels on the final random validation sample.

Stage / condition	n	Macro- F_1	Cohen’s κ
Step 3: Zero-shot, first development batch	250	.563	.415
Few-shot	250	.712	.638
Few-shot + CoT	250	.574	.472
Zero-shot, second development batch (new 250)	250	.849	.702
Step 4: Final zero-shot sample	600	.920	.839

Table 4: LLM agreement with the human consensus labels across development and final validation.

Figure 1 summarizes macro- F_1 and Cohen’s κ across the tested configurations. It illustrates the improvement across the development stages and the strong performance of the final zero-shot validation procedure. However, the configurations should not be compared without qualification because they were evaluated on different samples and, in Step 4, with a two-category validity evaluation rather than the original three-category evaluation.

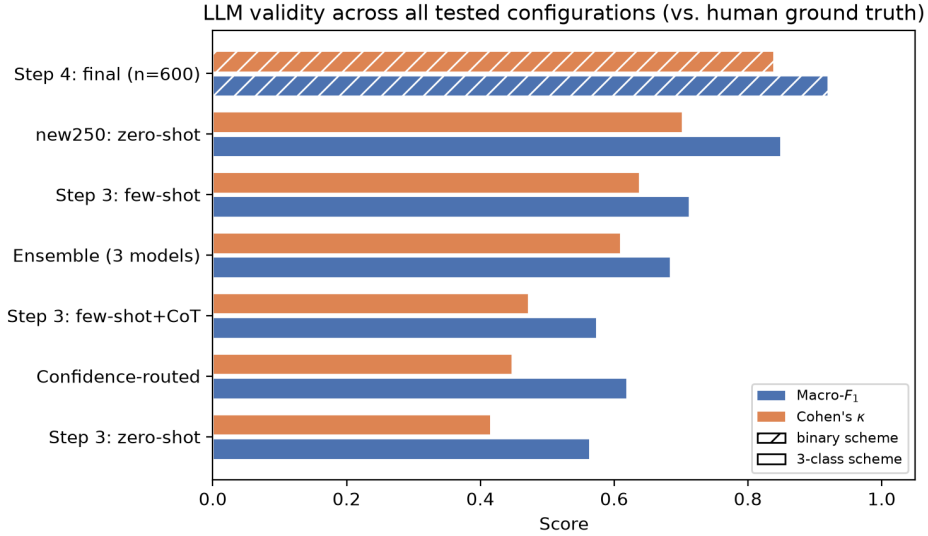


Figure 1: Macro- F_1 and Cohen’s κ across all tested configurations. Hatched bars represent the two-category validity evaluation in Step 4, while solid bars represent the three-category evaluation used during development.

5.3. Model Comparison: Ensembling vs. Confidence-Based Routing

Table 5 and Table 6 report the two side experiments described in Sections 4.2 and 4.2. On the development set used for these comparisons, multi-model ensembling with tie-break adjudication modestly improved performance relative to the corresponding single-model baseline, increasing macro- F_1 from .615 to .684 and Cohen’s κ from .593 to .610. Confidence-based routing, in contrast, did not improve performance and substantially reduced Cohen’s κ , because the confidence scores produced by the smaller model did not reliably distinguish correct from incorrect classifications.

Configuration	Macro- F_1	κ
Ministral-3-14b alone	0.615	0.593
Ensemble (+ Llama-3.3-70B, Qwen3.5-27b tie-break)	0.684	0.610

Table 5: Two-model ensemble vs. single-model annotation ($n = 250$; 74 tie-break cases, adjudicator sided with Model A on 61 and Model B on 13).

5.4. Substantive Framing Patterns Across Groups and Outlets

Beyond evaluating model validity, we used the final random sample for an exploratory comparison of negative crime framing across outlet categories. Following the application proposed by Freudenthaler, Ludwig, and Müller, we calculated the share of paragraphs

Configuration	Macro- F_1	κ
Ministral-3-14b alone (no routing)	0.618	0.601
Confidence-routed (244/250 rows routed)	0.619	0.447

Table 6: Confidence-based routing vs. single-model annotation ($n = 250$).

labeled `CRIME_FRAME_NEG` within each outlet category, that can be seen in Figure 2.

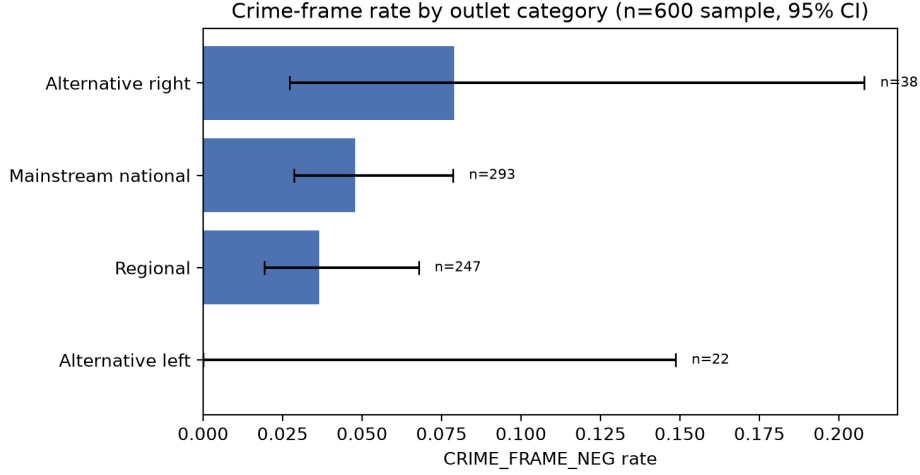


Figure 2: `CRIME_FRAME_NEG` rate by outlet category ($n = 600$)

Alternative-right outlets show the highest point estimate (7.9%, 3 of 38 paragraphs), roughly double the rate in mainstream (4.8%, 14 of 293) and regional (3.6%, 9 of 247) outlets, while no crime-frame paragraphs appeared among alternative-left outlets (0 of 22) in this sample. This pattern is directionally consistent with Wodak (2015)’s account of security-threat framing as a rhetorical strategy associated with right-wing populist discourse. However, the confidence intervals overlap substantially across all four categories, so this should be read as a descriptive pattern rather than a confirmed difference; a larger annotated sample, ideally covering the full 659,895-paragraph corpus, would be needed to test it with adequate statistical power.

6. Discussion

The repeated rounds of instruction development and hardcase review improved the clarity of the coding procedure for both the human annotators and the LLM. Recurring disagreements revealed where the initial instructions allowed different interpretations, particularly regarding the attribution of crime-related content to [Gruppe 1], the Germany-relatedness criterion, war reporting, and the distinction between victims and perpetrators. The increase in human intercoder reliability across the two development batches is consistent with a more shared understanding of these rules. However, it cannot be attributed solely to the revisions because the annotators also gained experience and the batches differed in their label distributions and types of cases. The consensus labels should therefore be understood as carefully adjudicated reference labels rather than as entirely objective ground truth.

The final random sample further illustrates why percentage agreement and Krippendorff’s α must be interpreted together. As reported in Section 5.1, we agreed on 95.5% of the paragraphs, but α was lower than in the second development batch. With only 26 cases of CRIME_FRAME_NEG, most agreement concerned the dominant NO_CRIME_FRAME category, while a small number of disagreements had a comparatively large effect on the chance-corrected reliability value. The remaining human hardcases mainly concerned the recently introduced Germany-relatedness criterion. Because the final sample was annotated almost one month after the second development batch, we may have become less accustomed to this rule and occasionally returned to earlier coding habits. This indicates that reliability depends not only on clear instructions but also on their continued application.

Figure 3 summarizes human reliability and LLM validity across the three stages. Human reliability increased between the development batches but was lower in the final random sample, partly because of the strongly imbalanced label distribution. LLM agreement with the human consensus labels was highest in the final validation. However, this does not mean that the LLM was more reliable than the human annotators, because Krippendorff’s α and Cohen’s κ measure different forms of agreement and were calculated on samples with different label distributions.

Changes in LLM performance across the instruction-development stages must also be interpreted with caution. The first 250-paragraph batch was repeatedly used to identify errors, revise the instructions, and select the few-shot examples. Performance on this batch may therefore partly reflect adaptation to its specific hardcases. The second batch had not been used in the initial prompting comparison described in Step 3.1, but it came from the same oversampled development sample and had a different label distribution. Because the instructions, prompting conditions, and sample composition differed across the development stages, the performance differences cannot be attributed to one specific change. The final random validation in Section 4.2 therefore provides the more relevant estimate of the finalized procedure for distinguishing NO_CRIME_FRAME from CRIME_FRAME_NEG.

The prompting comparison nevertheless produced a useful practical finding. Few-shot prompting performed better than zero-shot prompting on the development data,

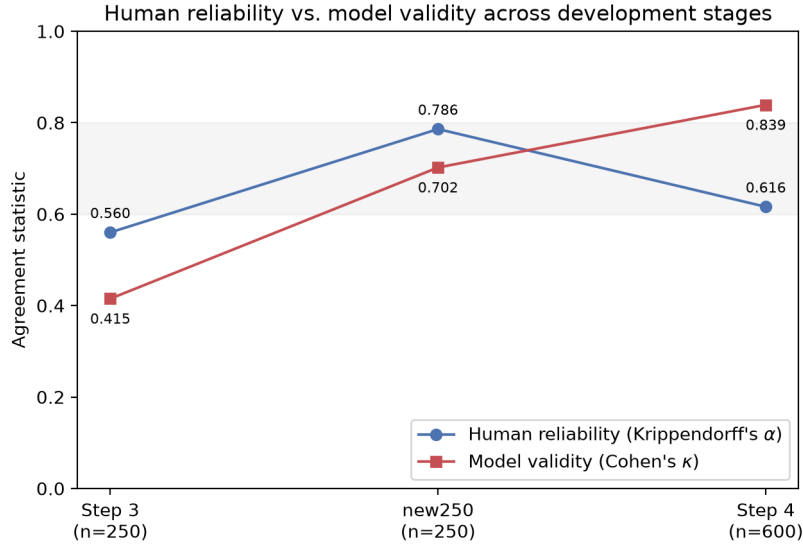


Figure 3: Human intercoder reliability and LLM validity across the development and validation stages. Krippendorff’s α measures agreement between the human annotators, whereas Cohen’s κ measures agreement between the LLM classifications and the human consensus labels.

suggesting that concrete examples helped the model apply the written rules to ambiguous paragraphs. Adding chain-of-thought explanations did not improve performance further. Concise examples therefore appeared more useful than longer explanations for this task. However, these prompting conditions were not tested again on the final random sample, so their relative performance cannot be generalized beyond the development data.

The results in Section 5.3 showed that confidence-based routing was not effective. The confidence scores did not reliably distinguish correct from incorrect classifications and caused nearly all paragraphs to be routed to the larger model. Disagreement between models was a more informative indicator of difficult cases, but the ensemble achieved only a limited improvement relative to its additional computational cost. The simpler single-model zero-shot procedure therefore offered a more suitable balance between validity, efficiency, and reproducibility. Its final classifications were based on ten repeated runs and majority voting, so the reported validity applies to the complete repeated-run procedure rather than to a single model response. The remaining LLM errors mainly concerned actor attribution and scope: the model generally recognized crime-related content but sometimes failed to determine whether [Gruppe 1] was presented as a perpetrator, victim, suspect, or only indirectly connected to an event.

Finally, the validated label captures only one dimension of group portrayal. A classification as **CRIME_FRAME_NEG** indicates that the focal group is associated with criminality, danger, or a security threat, but does not by itself establish racism in the broader theoretical sense discussed in Section 2. A paragraph without this frame may likewise contain

other stereotypical or discriminatory representations. The measure should therefore be interpreted as one component of the broader analytical framework rather than as a stand-alone assessment of a group’s overall portrayal in German news.

7. Limitations and Future Work

The instruction-development process relied on two relatively small, oversampled batches, and the first batch was used repeatedly for instruction revision and few-shot example selection. Although the final random validation reduced the risk of adaptation to the development data, it included only 600 paragraphs and 26 cases of `CRIME_FRAME_NEG`. Future research should evaluate the finalized procedure on a larger independent sample, including an oversampled set of negative crime-frame cases.

The final random sample was evaluated only with Minstral using the finalized zero-shot instructions, ten repeated runs, and majority voting. The few-shot, chain-of-thought, ensemble, and confidence-routing conditions were not tested again on this independent sample. Future studies should compare these prompting strategies and additional models on the same held-out test set.

The `CRIME_FRAME_POS` category could not be evaluated reliably because only one positive case remained in the development data after the instructions were refined. Targeted sampling is therefore needed to identify more positive security, prevention, and de-escalation frames and to assess this category separately.

The human reference labels were produced by the same two researchers who developed the instructions. Future validation should include an independent annotator and, where necessary, additional article context to support difficult decisions involving actor attribution and Germany-relatedness.

The outlet comparison was also limited by the small numbers of paragraphs from alternative-left and alternative-right outlets. Reliable comparisons across all 78 groups and four outlet categories require a larger stratified sample or application of the procedure to the full dataset.

Finally, the project examined only security and crime framing in German newspaper coverage from 2022. Future research should extend the procedure to other years, countries, and dimensions of the broader framework, particularly cultural and economic threat and benefit frames.

8. Conclusion

This project developed and documented an LLM-assisted procedure for identifying negative security and crime framing of ethnic, national, and religious groups in German news coverage. Through repeated instruction development, human annotation, and hardcase review, the final annotation instruction retained all three labels, while the validity evaluation focused on the distinction between `NO_CRIME_FRAME` and `CRIME_FRAME_NEG` because positive crime frames were too rare for reliable evaluation. On the independent random validation sample, the finalized zero-shot Minstral procedure with ten repeated

runs and majority voting achieved an accuracy of 98.7%, a macro- F_1 of .920, and Cohen's κ of .839. These results support the validity of the procedure while also showing the importance of interpreting performance under strong class imbalance.

Regarding the research question, the findings show that an LLM guided by iteratively refined annotation instructions can reliably and validly distinguish negative crime framing from paragraphs without such a frame. The comparison with the human annotators should nevertheless be interpreted cautiously because human reliability and LLM validity were assessed using different agreement measures. The exploratory outlet comparison in Figure 2 showed the highest negative crime-frame rate among alternative-right outlets, although the small number of cases and overlapping confidence intervals mean that this pattern remains descriptive.

The project further shows that clear decision rules, carefully reviewed examples, and qualitative analysis of disagreements are central to LLM-assisted content analysis. At the same time, the procedure remains limited by the small number of negative crime frames in the validation sample, the absence of a validated positive security frame, and its focus on one frame family, one country, and one year. The resulting measure should therefore be understood as one component of the broader framework proposed by Freudenthaler, Ludwig, and Müller, rather than as a complete assessment of racism or group portrayal. It nevertheless provides a basis for the broader goal of operationalizing ethnic and racial bias in news communication and enabling systematic comparisons of how different groups are portrayed across German news outlets. Future research should validate the procedure on larger and more diverse samples, compare additional models and prompting strategies, and extend the approach to cultural and economic frames.

Data and Code Availability

The raw news dataset used in this report was provided by the course instructor exclusively for academic use within this seminar and is not publicly available. However, to ensure transparency and reproducibility, all samples, raw human consensus labels, validation and reliability statistics, and complete data analysis scripts are publicly available on GitHub (<https://github.com/marmeeepandya/CAC-Project>). Besides that, the final annotation instructions in german language can be find in the Appendix A.1.

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A. Appendix

A.1. Annotation Instruction

The instruction below is the final version used in Step 4, after the revisions introduced during Steps 2 and 3, and is provided here in its original German formulation long with the corresponding reminder text.

System role: "Sie sind ein Experte für Inhaltsanalyse in einer Studie zur medialen Darstellung ethnischer und sozialer Gruppen in deutschen Nachrichten. Ihre Aufgabe ist es, zu klassifizieren, ob ein gegebener Nachrichtenabsatz einen Einwanderungs- und Kriminalitätsrahmen enthält und, falls ja, ob dieser negativ (Bedrohung) oder positiv (Sicherheitsgewinn/Prävention) dargestellt wird.

Sie erhalten Mehrfachabsätze aus deutschen Zeitungsartikeln. Es können mehrere Gruppen erwähnt werden. Gruppen können als [Gruppe 1] oder [andere Gruppe] bezeichnet werden. Kodieren Sie ausschließlich die Darstellung von [Gruppe 1]. Erwähnungen anderer Gruppen dienen lediglich als Kontext und dürfen Ihre Entscheidung nicht beeinflussen, sofern sie nicht ausdrücklich auf [Gruppe 1] bezogen werden. Konzentrieren Sie sich ausschließlich auf das, was explizit im Text steht. Schließen Sie keine Gruppenassoziationen, die nicht direkt ausgedrückt werden.

Der Kriminalitäts- oder Sicherheitsrahmen muss einen Bezug zu Deutschland haben. Ein Deutschlandbezug liegt auch vor, wenn Entscheidungen, Maßnahmen oder Bedrohungen auf Ebene der Europäischen Union oder der NATO beschrieben werden und Deutschland als Mitglied davon betroffen ist. Die bloße Erwähnung der EU oder NATO reicht nicht aus. Es muss erkennbar sein, dass die Entscheidung, Maßnahme oder Bedrohung auch Deutschland oder die gemeinsame Sicherheit der Mitgliedstaaten betrifft."

Instructions (decision rules):

NO_CRIME_FRAME wird vergeben, wenn kein expliziter Kriminalitäts- oder Sicherheitsrahmen vorliegt. Dies gilt insbesondere wenn: [Gruppe 1] gar nicht erwähnt wird; Kriminalität erwähnt wird, aber nicht mit [Gruppe 1] verknüpft ist; Ausschließlich eine andere Gruppe als [Gruppe 1] mit Kriminalität oder Sicherheit verknüpft wird; [Gruppe 1] als Opfer von Straftaten dargestellt wird und kein kriminelles Verhalten zeigt; Kriminalität auf soziale Bedingungen (Armut, Ausgrenzung etc.) zurückgeführt wird; Der Text Kriminalitätszuschreibungen relativiert oder widerspricht („kein Generalverdacht“, „nicht alle“); Krieg, geopolitische Konflikte oder militärische Ereignisse beschrieben werden; Der Einsatz von Bomben, Luftangriffen, Raketen, Drohnen, Atomwaffen oder nuklearen Waffen im Kontext von Krieg, Militär, geopolitischen Konflikten oder staatlicher Kriegsführung beschrieben wird; Humanitäre Krisen oder Naturkatastrophen beschrieben werden; Ein Täter erwähnt wird, aber NICHT Mitglied von [Gruppe 1] ist; Fiktionale, kulturelle oder satirische Darstellungen beschrieben werden, z.B. TV-Shows, Filme, Serien, Romane, Theaterstücke, Comics, Karikaturen oder Kunstwerke, solange [Gruppe 1] nicht als reale kriminelle oder sicherheitsgefährdende Akteurin dargestellt wird; Demonstrationen, Proteste, Kundgebungen oder politische Versammlungen erwähnt werden, aber nicht mit Kriminalität, Gewalt, Sicherheitsbedrohung, Prävention oder konkreten sicherheitsbezogenen Maßnahmen verknüpft sind; Wenn Kriminalität,

Gewalt oder Bedrohung nur als Propaganda, falsches Narrativ, Diffamierung oder unbelegte Zuschreibung gegen [Gruppe 1] erwähnt wird und der Text diese Darstellung als manipulativ oder falsch einordnet. In solchen Fällen wird [Gruppe 1] als Ziel oder Opfer der Zuschreibung dargestellt, nicht als tatsächlicher Täter oder Sicherheitsrisiko; Wenn ein Ereignis ausschließlich im Ausland stattfindet und weder Deutschland noch deutsche Behörden, deutsche Institutionen, deutsche Staatsangehörige oder die Sicherheit Deutschlands ausdrücklich betroffen sind. Ein Bezug zur EU oder NATO zählt nur, wenn Deutschland als Mitglied von der beschriebenen Entscheidung, Maßnahme oder Bedrohung betroffen ist; Wenn die EU oder NATO nur beiläufig erwähnt wird, ohne dass eine konkrete gemeinsame Sicherheitsentscheidung, Maßnahme oder Bedrohung beschrieben wird, die Deutschland als Mitglied betrifft.

WICHTIG: NO_CRIME_FRAME nur vergeben, wenn [Gruppe 1] eindeutig als Opfer dargestellt wird und nicht selbst mit kriminell Verhalten, Verdacht, Ermittlungen, Gewaltunterstützung oder Sicherheitsgefahr in Verbindung steht. Wenn ein Mitglied von [Gruppe 1] eine Straftat begeht, mutmaßlich begeht, verdächtigt wird oder daran beteiligt ist, bleibt dies CRIME_FRAME_NEG – auch dann, wenn die Handlung durch äußeren Druck, Anreize oder vermeintliche Vorteile (z.B. schnellerer Erhalt von Papieren) motiviert ist, sofern zusätzlich ein Bezug zu Deutschland besteht.

CRIME_FRAME_NEG wird vergeben, wenn [Gruppe 1] als Quelle von Kriminalität oder Sicherheitsbedrohung dargestellt wird. Dies umfasst: [Gruppe 1] oder ein Mitglied von [Gruppe 1] wird als Täter, Verdächtiger, Beschuldigter oder kriminell dargestellt; [Gruppe 1] oder ein Mitglied von [Gruppe 1] wird mit konkreten Straftaten verbunden (z.B. Diebstahl, Gewalt, Mord); Asylmissbrauch oder erschlundene Asylleistungen, sofern [Gruppe 1] oder ein Mitglied von [Gruppe 1] ausdrücklich mit strafbarem oder betrügerischem Verhalten verbunden wird; [Gruppe 1] wird als Ursache steigender Kriminalität dargestellt; [Gruppe 1] wird als Gefahr für öffentliche oder nationale Sicherheit dargestellt; [Gruppe 1] wird als Gefahr für die Sicherheit Deutschlands oder der Europäischen Union dargestellt; Migration oder die Zugehörigkeit zu [Gruppe 1] wird mit Kriminalität verknüpft; Terrorismus oder Extremismus wird mit [Gruppe 1] verbunden; Physische Gewalt oder Bedrohung geht explizit von Mitgliedern von [Gruppe 1] aus; [Gruppe 1] begrüßt, legitimiert, feiert, rechtfertigt oder unterstützt einen Anschlag, Terrorakt, Gewaltakt oder eine Straftat; [Gruppe 1] stellt einen Anschlag, Terrorakt, Gewaltakt oder eine Straftat als Widerstand, heroische Operation oder legitime Handlung dar; [Gruppe 1] erscheint als Unterstützer, Kommentator, ideologischer Bezugspunkt oder politischer Akteur, der explizit mit Zustimmung, Rechtfertigung, Feier, Unterstützung oder ideologischer Nähe zu Terrorismus, politischer Gewalt oder Straftaten verbunden wird; Mutmaßungen, Verdachtsfälle, Ermittlungen, Beschuldigungen oder noch nicht gerichtlich bestätigte Vorwürfe zählen als CRIME_FRAME_NEG, wenn [Gruppe 1] oder ein Mitglied von [Gruppe 1] explizit mit der mutmaßlichen Straftat, dem Verdacht oder den Ermittlungen verknüpft wird (z.B. „mutmaßlich“, „verdächtigt“, „beschuldigt“, „soll getan haben“, „laut Polizei“, „Ermittlungen gegen“, „unter Verdacht“); Polizeiliche Maßnahmen gegen ein Mitglied von [Gruppe 1], z.B. Festnahme, Durchsuchung, Fahndung, Ermittlungen, Drogentest oder Anklage, wenn das Gruppenmitglied als Täter, Verdächtiger oder Gesuchter erscheint; Eine deutsche Behörde, Institution oder Amtsperson stellt [Gruppe 1] ausdrücklich als Kriminalitätsproblem oder Sicherheitsrisiko für Deutschland dar; [Gruppe 1] oder Migration wird als Ursache von Gewalt, Chaos,

Kontrollverlust oder gesellschaftlicher Bedrohung in Deutschland dargestellt; Ein Mitglied von [Gruppe 1] übt in Deutschland Zwang, Drohung oder autoritäre Kontrolle aus (z.B. Kopftuchzwang an Schulen); Eine Straftat oder ein Anschlag im Ausland wird mit Deutschland verbunden, z.B. weil der Täter oder das Opfer aus Deutschland kommt, in Deutschland lebt oder lebte, die deutsche Staatsangehörigkeit besitzt, deutsche Behörden beteiligt sind oder deutsche Ziele betroffen sind; Eine Institution der EU oder NATO stellt [Gruppe 1] als Sicherheitsrisiko dar oder beschließt eine sicherheitsbezogene Maßnahme, die Deutschland als Mitglied betrifft; Eine Bedrohung richtet sich gegen die EU, die NATO oder deren Mitgliedstaaten und betrifft damit auch die gemeinsame Sicherheit Deutschlands.

Achtung: Die Bezeichnung „illegale Einwanderer“ fällt nicht unter diesen Topos, wenn nicht noch weitere Verbrechen genannt werden. Schleuser aus [Gruppe 1], die als Kriminelle dargestellt werden, zählen aber als CRIME_FRAME_NEG.

CRIME_FRAME_POS wird vergeben, wenn ein Sicherheits- oder Kriminalitätsrahmen vorhanden ist, aber positiv dargestellt wird. Dies umfasst: Maßnahmen zur Kriminalitätsbekämpfung im Zusammenhang mit [Gruppe 1]; Prävention im Zusammenhang mit [Gruppe 1] (z.B. soziale Arbeit, Integration, Deradikalisierung); Differenzierung in Bezug auf [Gruppe 1] („kein Generalverdacht“, „nicht alle“); Kriminalität wird durch soziale Faktoren erklärt statt durch [Gruppe 1] selbst; Mitglieder von [Gruppe 1] wirken aktiv an Sicherheit oder Prävention mit (z.B. Polizei, Sozialarbeit); Maßnahmen werden als Verbesserung von Sicherheit oder Ordnung dargestellt; Maßnahmen der EU oder NATO gelten ebenfalls als CRIME_FRAME_POS, wenn sie ausdrücklich der Sicherheit Deutschlands oder der gemeinsamen Sicherheit der Mitgliedstaaten dienen; Demonstrationen, Proteste oder Kundgebungen sind nur dann CRIME_FRAME_POS, wenn sie ausdrücklich zu konkreten gesetzlichen, politischen, polizeilichen oder präventiven Änderungen führen oder als Beitrag zu mehr Sicherheit, Schutz, Prävention oder Rechtsstaatlichkeit dargestellt werden.

Abgrenzung: Polizeiliche Maßnahmen gegen ein Mitglied von [Gruppe 1] sind CRIME_FRAME_NEG, wenn das Gruppenmitglied als Täter oder Verdächtiger erscheint. CRIME_FRAME_POS wird nur vergeben, wenn Schutz, Prävention, Differenzierung oder Sicherheitsgewinn im Vordergrund steht. Präventions- oder Schutzmaßnahmen im Ausland zählen nur dann, wenn ein ausdrücklicher Bezug zu Deutschland oder zu einer gemeinsamen Sicherheitsmaßnahme der EU oder NATO besteht.

Reminder: Vergeben Sie genau ein Label: NO_CRIME_FRAME, CRIME_FRAME_NEG oder CRIME_FRAME_POS. Achten Sie besonders darauf, dass die Gruppe explizit mit Kriminalität, Sicherheit, Terrorismus, Gewaltlegitimation, Verdacht, Ermittlungen oder Prävention verknüpft ist. Wenn Kriminalität erwähnt wird, aber nicht mit der Gruppe verbunden ist oder die Gruppe Opfer ist, vergeben Sie NO_CRIME_FRAME. Mutmaßungen, Verdacht oder Ermittlungen zählen als CRIME_FRAME_NEG, wenn die Gruppe explizit mit der mutmaßlichen Straftat verbunden wird. Kriegskontexte, fiktionale Inhalte und bloße Demonstrationen ohne Sicherheits- oder Präventionsbezug sind NO_CRIME_FRAME.

A.2. Few-Shot Examples Used in the Instruction Revision (Step 3.1)

The 13 few-shot examples used in the instruction-revision round (Section 4.2), each paired with its label.

- **NO_CRIME_FRAME**: "Die Lage in der Unterkunft hat sich nach Angaben der Sozialarbeiter entspannt. Viele [Gruppe 1] hätten inzwischen Deutschkurse begonnen und bemühten sich aktiv um eine Arbeitsstelle. Die Caritas lobte das Engagement der Freiwilligen, die regelmäßig Unterstützung anbieten."
- **CRIME_FRAME_NEG**: "Ein 34-jähriger [Gruppe 1] wurde am Freitagabend am Hauptbahnhof festgenommen. Er soll in den vergangenen Wochen mehrfach Passanten mit einem Messer bedroht und Handtaschen entrissen haben. Die Polizei hat ihn als Serientäter eingestuft."
- **CRIME_FRAME_NEG**: "Die Innenministerkonferenz hat beschlossen, die erkennungsdienstliche Erfassung aller einreisenden [Gruppe 1] zu verschärfen. Innenminister Hoffmann erklärte, man könne sich keine Sicherheitslücken leisten."
- **NO_CRIME_FRAME**: "Auf dem Gelände der Flüchtlingsunterkunft in Tempelhof wurden in der Nacht Fensterscheiben eingeworfen und Hakenkreuze gesprüht. Die [Gruppe 1] Bewohner wurden unverletzt, aber stark verängstigt in eine Notunterkunft gebracht."
- **NO_CRIME_FRAME**: "Die [Gruppe 1] Streitkräfte haben nach Angaben des Verteidigungsministeriums in Kiew erneut mehrere Städte im Osten beschossen. Die NATO beriet über weitere Waffenlieferungen."
- **CRIME_FRAME_NEG**: "Sprecher der [Gruppe 1] bezeichneten den Anschlag als legitimen Widerstand und erklärten, die Opfer seien selbst schuld."
- **CRIME_FRAME_POS**: "Das Gemeinsame Terrorismusabwehrzentrum hat nach intensiver Zusammenarbeit von Polizei und Geheimdiensten einen islamistischen Anschlag in München vereitelt."
- **CRIME_FRAME_POS**: "Experten erklären den Anstieg der Jugendkriminalität in Brennpunktvierteln vor allem mit sozialer Ausgrenzung und fehlenden Perspektiven für [Gruppe 1] Jugendliche."
- **CRIME_FRAME_POS**: "Das neue Deradikalisierungsprogramm der Bundesregierung soll vor allem [Gruppe 1] Jugendliche ansprechen, die gefährdet sind, in extremistische Kreise abzuweichen."
- **NO_CRIME_FRAME**: "Die Stadtratsfraktion diskutierte Maßnahmen zur besseren Integration von [Gruppe 1]. Ein konkreter Sicherheits- oder Kriminalitätsbezug wurde nicht hergestellt."
- **CRIME_FRAME_NEG**: "Die Behörden berichten, dass sich in Teilen der Stadt regelrechte Parallelgesellschaften gebildet haben, in denen [Gruppe 1] unter sich bleiben und staatliche Kontrolle kaum noch greift."
- **CRIME_FRAME_NEG**: "Der Botschafter warnte: „Diese [Gruppe 1] stellen ein Sicherheitsrisiko für Deutschland dar.“"
- **NO_CRIME_FRAME**: "Die Politikerin forderte eine bessere Integrationspolitik für [Gruppe 1]. Fragen der öffentlichen Ordnung wurden angesprochen, ohne konkrete Vorfälle oder Straftaten zu nennen."

A.3. Example of a Positive Crime Frame Classification

The following paragraph from the dataset provides a representative example of a text classified as `CRIME_FRAME_POS`. Here, a crime- and security-related frame is explicitly established through the mention of human trafficking prevention, police measures, and criminal investigators, while [Gruppe 1] is framed protectively as a vulnerable group that needs to be shielded from exploitation.

Ehrenamtliche Helfer:innen kritisieren, dass keine Vermittlung stattfindet und deren Expertise nicht berücksichtigt werde. Die Sozialverwaltung weist dies zurück und sagt, Repräsentanten der Ehrenamtlichen seien in regelmäßige, wöchentliche Gespräche einbezogen.

Karuna-Vorstand Jörg Richert erklärte dem Tagesspiegel, man habe am Hauptbahnhof wegen des großen Ansturms keine Vermittlung zwischen Privatleuten und [Gruppe 1] mehr durchführen können. Künftig werde es ein nachprüfbares Verfahren geben. Karuna werde die Vermittlung von Plätzen in Tegel wieder aufnehmen. Zunächst sollen vulnerable Gruppen wie behinderte oder pflegebedürftige Menschen sowie Personen, deren Weiterreise erst Tage später stattfinden kann und Familienmitglieder, die auf der Flucht getrennt wurden, vermittelt werden.

Um [Gruppe 1] vor Missbrauch bei privaten Unterbringungsangeboten zu schützen, setzt die Polizei am Hauptbahnhof, dem Zentralen Omnibusbahnhof und im Ankunftszentrum auch verdeckte Ermittler ein. Eine für Kinderschutz zuständige Ermittlungsgruppe des Landeskriminalamtes ist dort ebenfalls im Einsatz.

A.4. Example of an LLM Hardcase: Propaganda and Contested Accusations

This LLM hardcase was identified during Step 2 and led to the need of an extra rule, the propaganda rule. Although the paragraph contains explicit terrorism rhetoric, the report presents these claims as a contested account by a state-affiliated source rather than as an established description of [Gruppe 1]. The correct label was therefore `NO_CRIME_FRAME`.

Die regierungsnahe Nachrichtenagentur Tasnim stellte die Situation anders dar: In der Nacht zum Sonntag hätten „bewaffnete Terroristen“ Privathäuser und öffentliche Einrichtungen in Brand gesetzt und die ganze Stadt und deren Einwohner in Panik versetzt. Mehrere Anführer der „Terrorgruppen“ hätten jedoch überführt und inhaftiert werden können, so der Tasnim-Bericht unter Berufung auf örtliche Sicherheitsbehörden.

Medienberichten zufolge gab es am Samstagabend auch in anderen Teilen des Landes erneut Proteste gegen den repressiven Kurs der islamischen Führung. Die in Norwegen ansässige Menschenrechtsorganisation Hengaw wusste am Sonnabend von einer „kritischen“ Lage in der Stadt Diwandarreh in der westlichen Provinz Kurdistan zu berichten, wo Regierungstruppen mindestens drei Zivilisten erschossen hätten. Am Sonntag äußerte sich

Hengaw zudem besorgt über die Lage in anderen mehrheitlich von [Gruppe 1] bewohnten Städten, darunter Bukan und Sakes. Dort hatten die Proteste zuletzt zugenommen.

A.5. Example of an LLM Hardcase: Actor-Attribution Error

In this LLM hardcase of the test batch (n=600), the model assigned `CRIME_FRAME_NEG`, whereas the human consensus label was `NO_CRIME_FRAME`. Although [Gruppe 1] appears close to crime-related terms, the group is connected to the victim's living environment or is the victim itself, as we as human annotators stated, rather than the perpetrator. The model error therefore resulted from incorrect actor attribution.

„Das ist kein schönes Erlebnis“, sagt er am Samstagmorgen, während er eine rote Grabkerze am Tatort abstellt, auf die seine Mutter eine Minute zuvor „Ruhe in Frieden“ geschrieben hat. Die bunten Rosen, die sie mitgebracht haben, stehen in einem Glas mit Wasser. „Das war gestern ein Scheißtag“, sagt Gino mit Tränen in den Augen. [Gruppe 1] wohnte 200 Meter vom Tatort entfernt Die Frau erlitt bei dem Angriff lebensgefährliche Stichverletzungen und starb noch vor Ort.

Der mutmaßliche Täter flüchtete nach der Tat und erschien drei Stunden später, gegen 13 Uhr, mit Mitarbeitern einer 200 Meter entfernt liegenden Geflüchtetenunterkunft, in der die [Gruppe 1] und ihre Kinder gelebt hatten, am Tatort. Dort wurde er von Zeugen erkannt und von der Polizei festgenommen.

A.6. Annotation Label Scheme

Label	Definition
NO_CRIME_FRAME	No explicit link between the group and crime, security, danger, violence, terrorism, suspicion, investigation, or prevention; includes victim portrayals, war/geopolitical reporting, and fictional depictions.
CRIME_FRAME_NEG	Group explicitly portrayed as source of crime, danger, or security threat (perpetrator, suspect, accused, endorser of violence).
CRIME_FRAME_POS	Security-crime frame present but positively valenced: the group actively contributes to safety, protection, prevention, policing, or de-escalation.

Table 7: Summary of the final annotation label scheme (Appendix A.1 gives the full instructions).

Authorship Allocation

Authorship of the sections was split up as follows:

1. **Abstract** – Robin-Kiara Braun & Marmee Pandya
2. **Introduction** – Marmee Pandya & Robin-Kiara Braun
3. **Theoretical Background** – Marmee Pandya
4. **Data** – Robin-Kiara Braun
5. **Method**
 - (a) **Sampling Strategy** – Robin-Kiara Braun
 - (b) **Annotation Instruction Steps** – Robin-Kiara Braun & (Step 3: Marmee Pandya)
6. **Results**
 - (a) **Human Intercoder Reliability** – Robin-Kiara Braun
 - (b) **LLM Validity Against Human Ground Truth** – Robin-Kiara Braun & Marmee Pandya
 - (c) **Model Comparison: Ensembling vs. Confidence-Based Routing** – Marmee Pandya
 - (d) **Substantive Framing Patterns Across Groups and Outlets** – Marmee Pandya
7. **Discussion** – Robin-Kiara Braun & Marmee Pandya
8. **Limitations and Future Work** – Robin-Kiara Braun
9. **Conclusion** – Robin-Kiara Braun & Marmee Pandya

Regarding the practical development, the annotation instructions were iteratively adapted by both. Specifically, Step 2 and 3 were developed by Robin-Kiara Braun, whereas Marmee Pandya worked on the few Shot and Chain of thoughts analysis, and Step 4 was jointly implemented by both authors. Furthermore, we collaboratively performed the human labeling process, manually annotating all samples to establish the dataset ground truth.

Use of AI Tools

Declaration

Task	Tool	Where?
Outline and structure planning	ChatGPT (5.5)	Before start writing
Broaden understanding of papers and concepts	NotebookLM (Audio), ChatGPT (5.5)	Throughout, refreshing knowledge
Translation of paragraphs and instructions	DeepL, ChatGPT (5.5)	Throughout coding process
Grammar, transitions, and academic phrasing	ChatGPT (5.5), DeepL	Throughout
BibTeX and LaTeX support (e.g., citations, figures)	Claude, Gemini	LaTeX formatting / Setup phase
Template understanding and formatting	Claude, Gemini	Setup phase
Coding and instruction development	Claude, ChatGPT (5.5)	Analysis phase

Statutory Declaration

I hereby guarantee that the present paper is my own and that I have properly acknowledged any help or support received from other individuals. Furthermore, I declare that neither this work nor parts of it have been submitted by myself or others to another university as part of a formal degree requirement. All intellectual property of others is clearly cited as such. All secondary literature and other sources are certified and listed in the bibliography. The same applies for graphic illustrations, pictures and all internet sources. I agree that my work may be electronically saved and sent anonymized to be checked for plagiarism. I am aware that this paper cannot be graded if this declaration is not signed.

Mannheim, July 21, 2026



Robin-Kiara Braun



Marmee Pandya